SpMV–G2 / SpMV: measured locality and bottleneck diagnosis

Weight: 10% of the course grade | Due date: to be announced

Experimental goal

Identify where time is spent and which memory-level hypotheses remain plausible on the two-socket Xeon Gold 6526Y server (per-core L2 ≈2 MiB; shared L3 ≈37.5 MiB/socket). Benchmark is single-threaded and first-touch; the sweep varies x from 0.5 to 64 MiB. Do NOT assume permuted must have more generic or LLC misses.

Required experiments and evidence

Step	Must do / interpretation	Marks
1. Freeze + validate	Run self-test, document compiler/source hashes, CPU affinity, memory topology, dimensions and CORRECT=PASS.	2
2. Paired measurement	≥5 runs of local/permuted per size; median ms/SpMV, ns/NNZ, cycles and IPC where available; plot x-size versus time/NNZ.	2
3. Responsible code	Capture perf record/report identifying csr_spmv; annotate hot lines/instructions if supported.	2
4. Memory hierarchy	Report supported kernel-only L1D/L2/LLC/dTLB events or N/A with reasons; relate misses to the cache-footprint model.	2
5. Defensible result	Give observation, most likely cause, alternative cause, a discriminating test and confidence; account for PMU scope and noise.	2

What the profiler can and cannot prove

Internal --pmu counts cycles, instructions and generic misses; --pmu-levels counts supported L1D load, L2 demand-read, LLC load and dTLB load misses. Both exclude setup/warm-up/verification, but are separate runs. N/A means unavailable; low PMU running_fraction invalidates raw counts. Neither generic misses nor LLC misses directly equals DRAM traffic.

Required size design

R	x payload	Reason to include
16,384	0.5 MiB	Smaller than one core's L2
65,536	2 MiB	Near one core's L2
262,144	8 MiB	Beyond L2; below socket LLC
1,048,576	32 MiB	Near socket LLC capacity
2,097,152	64 MiB	Above socket LLC capacity

Cache thresholds are hypotheses, not guaranteed transitions: CSR arrays and other processes compete for capacity; CPU frequency, prefetchers, page translation and NUMA also matter.

Exact G2 commands (copy inside the extracted package directory)

```
g++ -O3 -g -fno-omit-frame-pointer -std=c++17 -Wall -Wextra \
  SpMV_Guided_Benchmark.cpp -o spmv_bench
./spmv_bench --selftest
python3 Run_SpMV_Experiments.py --reps 5
python3 Plot_SpMV_Results.py results/summary.csv
cat results/summary.csv
```

The runner pins one CPU, executes separate core and hierarchy counter passes for every case, and saves both logs plus raw.csv, summary.csv and provenance.txt. It refuses to overwrite an existing result directory.

Hotspot and cache-level investigation

```
perf list | grep -Ei "L1-dcache|LLC-load|dTLB-load|l2_rqsts"
perf record -g -e cycles:u -o results/profile.data -- \
  ./spmv_bench --pattern permuted --rows 1048576 --nnz-per-row 8 --iters 60
perf report -i results/profile.data --stdio > results/hotspots.txt
```

Kernel-only counter commands (additional examples; included in the automated runner)

```
./spmv_bench --pmu --pattern local --rows 1048576 --nnz-per-row 8 --iters 60
./spmv_bench --pmu-levels --pattern local --rows 1048576 --nnz-per-row 8 --iters 60
# Repeat both commands with --pattern permuted
```

L2 uses Intel Emerald Rapids event L2_RQSTS.DEMAND_DATA_RD_MISS only on family 6/model 207. It excludes merged requests and prefetch misses; counters are not additive. If any event is unsupported or the count is multiplexed, report N/A and retain the reason in logs.

Submit (SpMV_G2_<studentID>)

report.pdf (≤4 pages): controlled-size plot, 5-run variability, cache footprint model, hotspot excerpt and a falsifiable diagnosis. Attach results/ in full, including raw logs and provenance; attach hotspots.txt or a documented perf failure. No compulsory DRAM/uncore, storage, sudo or hardware accelerator measurements.